SOF NSO Science — Topic Deep-Dive: Matter & Materials

For Ambar, Class III CBSE

---

1. Three states of matter

Everything around us is made of **matter**. Matter is found in three forms called **states**: **solid, liquid and gas**.

State	Shape	Can it be poured?	Examples
Solid	Fixed shape of its own	No	Stone, chair, book, ice, iron nail
Liquid	Takes the shape of its container	Yes	Water, milk, oil, honey, juice
Gas	No fixed shape; fills all the space	Yes (flows)	Air, steam, the gas in a balloon

Key ideas: - A **solid** keeps its own shape. You must press or cut it to change it (like clay). - A **liquid** flows and takes the shape of the glass, bottle or jar it is in. - A **gas** spreads out to fill the whole container. That is why a balloon becomes big when we fill it with air.

Trap: Honey is thick and sticky, but it **flows** and takes the shape of its jar — so honey is a **liquid**, not a solid.

---

2. Natural and man-made materials

Type	Meaning	Examples
Natural	Found ready in nature	Wood (trees), cotton (plant), wool (sheep), stone, water
Man-made	Made by people in factories	Plastic, glass, nylon, paper

Cotton and wool grow in nature; **plastic and nylon** are made by people.

---

3. Uses of materials

We choose a material for a job because of its special property.

Material	Property	We use it to make
Wood	Strong, does not let heat pass	Tables, chairs, doors
Metal	Strong, lasts long, does not burn	Spoons, vessels, tools
Plastic	Light, does not let water through	Buckets, bottles, raincoats
Glass	See-through (transparent)	Windows, tumblers, mirrors
Rubber	Can stretch and bend	Tyres, balls, erasers
Cloth	Soft	Clothes, curtains, bedsheets

We make cooking pots of **metal** (not paper) because metal is strong and does not burn. We make windows of **glass** so that light can pass and we can see through.

---

4. Floating and sinking

Put things in water and some **float** while some **sink**. - **Float:** wooden stick, dry leaf, plastic ball, empty bottle. - **Sink:** iron nail, stone, coin, key.

Different materials behave differently. Interestingly, a heavy iron ship floats because of its **shape** — so shape can help heavy things float.

---

5. Soluble and insoluble

When we mix a substance in water: - **Soluble** = it dissolves and disappears into the water. Examples: **salt, sugar**. - **Insoluble** = it does not dissolve; it settles at the bottom. Examples: **sand, chalk powder, stones**.

If we leave salt water in the sun, the water evaporates and the **salt is left behind** — showing the salt was still there, just dissolved.

Common traps

The trap	The truth
"Honey is a solid because it is thick"	Honey **flows** → it is a **liquid**
"Plastic and glass are natural"	They are **man-made**
"Cotton is man-made"	Cotton is **natural** (from a plant)
"Sand dissolves in water"	Sand is **insoluble**; it settles down
"All heavy things sink"	A heavy iron ship **floats** because of its shape
"A gas has a fixed shape"	A gas has **no fixed shape**; it fills all the space

High-frequency question types

1. **State of matter:** "Air is a ___" → gas.
2. **Property:** "A liquid takes the shape of its ___" → container.
3. **Natural vs man-made:** "Which is man-made?" → plastic.
4. **Material use:** "Tyres are made of ___" → rubber.
5. **Float/sink:** "Which floats?" → wooden stick.
6. **Soluble:** "Which dissolves in water?" → sugar/salt.

Key facts checklist

- ☐ Three states: solid, liquid, gas.
- ☐ Solid = fixed shape; liquid = takes container shape; gas = fills all space.
- ☐ Honey is a liquid; ice is solid water; steam is gas.
- ☐ Natural: wood, cotton, wool, stone. Man-made: plastic, glass, nylon.
- ☐ Wood → furniture; metal → vessels; glass → windows; rubber → tyres; cloth → clothes.
- ☐ Wood floats; iron nail and stone sink.
- ☐ Salt and sugar dissolve (soluble); sand does not (insoluble).

Quick drill (10 questions)

1. Name the three states of matter.
2. Which state takes the shape of its container?
3. Is honey a solid or a liquid?
4. Is plastic natural or man-made?

5. Tyres are made of which material?
6. Windows are made of which see-through material?
7. Does a wooden stick float or sink in water?
8. Does an iron nail float or sink?
9. Which dissolves in water: sugar or sand?
10. Ice is the ___ form of water.

Answers: 1. Solid, liquid, gas 2. Liquid 3. Liquid 4. Man-made 5. Rubber 6. Glass 7. Float 8. Sink 9. Sugar 10. Solid